

PRACTICAL COURSE WORKBOOK

Data Warehousing & BigQuery

Model, query and explain dependable data

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a working data model with validated queries
SKILLS	BigQuery, Snowflake, dbt, Data Warehouse, ETL, Critical evaluation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use BigQuery appropriately in a realistic, bounded task.
- Use Snowflake appropriately in a realistic, bounded task.
- Use dbt appropriately in a realistic, bounded task.
- Use Data Warehouse appropriately in a realistic, bounded task.

WHO THIS IS FOR

Developers and analysts who need dependable data systems.

BEFORE YOU BEGIN

- Basic programming or spreadsheet experience

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

Data Warehousing & BigQuery is a structured, practical course covering BigQuery, Snowflake, dbt, Data Warehouse, ETL. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 BigQuery: Data modelling	1. Entities with BigQuery 2. Relationships with Snowflake 3. Constraints with dbt
02 Snowflake: Querying	1. Selection with Data Warehouse 2. Joins with ETL 3. Aggregation with BigQuery
03 dbt: Reliability	1. Transactions with Snowflake 2. Indexes with dbt 3. Backups with Data Warehouse
04 Data Warehouse: Production design	1. Security with ETL 2. Performance with BigQuery 3. Operations with Snowflake

HOW TO USE EACH LESSON PAGE

- 1 Read the objective and identify the decision it supports.
- 2 Attempt the retrieval prompt before checking the online explanation.
- 3 Reproduce the worked example with the stated input.
- 4 Test one normal case and one boundary or failure case.
- 5 Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

BigQuery: Data modelling

Apply BigQuery through entities, relationships, constraints.

LESSON 01 / 18 MIN

Entities with BigQuery

Learning objectives

- Explain entities in the context of Data Warehousing & BigQuery.
- Apply BigQuery to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

entities, BigQuery, databases

Retrieval prompt

In Data Warehousing & BigQuery, which evidence best supports a entities result produced with BigQuery?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 02 / 22 MIN

Relationships with Snowflake

Learning objectives

- Explain relationships in the context of Data Warehousing & BigQuery.
- Apply Snowflake to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

relationships, Snowflake, databases

Retrieval prompt

In Data Warehousing & BigQuery, which evidence best supports a relationships result produced with Snowflake?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 03 / 26 MIN

Constraints with dbt

Learning objectives

- Explain constraints in the context of Data Warehousing & BigQuery.
- Apply dbt to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

constraints, dbt, databases

Retrieval prompt

In Data Warehousing & BigQuery, which evidence best supports a constraints result produced with dbt?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

Snowflake: Querying

Apply Snowflake through selection, joins, aggregation.

LESSON 04 / 30 MIN

Selection with Data Warehouse

Learning objectives

- Explain selection in the context of Data Warehousing & BigQuery.
- Apply Data Warehouse to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

selection, Data Warehouse, databases

Retrieval prompt

In Data Warehousing & BigQuery, which evidence best supports a selection result produced with Data Warehouse?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 05 / 18 MIN

Joins with ETL

Learning objectives

- Explain joins in the context of Data Warehousing & BigQuery.
- Apply ETL to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

joins, ETL, databases

Retrieval prompt

In Data Warehousing & BigQuery, which evidence best supports a joins result produced with ETL?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 06 / 22 MIN

Aggregation with BigQuery

Learning objectives

- Explain aggregation in the context of Data Warehousing & BigQuery.
- Apply BigQuery to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

aggregation, BigQuery, databases

Retrieval prompt

In Data Warehousing & BigQuery, which evidence best supports a aggregation result produced with BigQuery?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

dbt: Reliability

Apply dbt through transactions, indexes, backups.

LESSON 07 / 26 MIN

Transactions with Snowflake

Learning objectives

- Explain transactions in the context of Data Warehousing & BigQuery.
- Apply Snowflake to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

transactions, Snowflake, databases

Retrieval prompt

In Data Warehousing & BigQuery, which evidence best supports a transactions result produced with Snowflake?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 08 / 30 MIN

Indexes with dbt

Learning objectives

- Explain indexes in the context of Data Warehousing & BigQuery.
- Apply dbt to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

indexes, dbt, databases

Retrieval prompt

In Data Warehousing & BigQuery, which evidence best supports a indexes result produced with dbt?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 09 / 18 MIN

Backups with Data Warehouse

Learning objectives

- Explain backups in the context of Data Warehousing & BigQuery.
- Apply Data Warehouse to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

backups, Data Warehouse, databases

Retrieval prompt

In Data Warehousing & BigQuery, which evidence best supports a backups result produced with Data Warehouse?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

Data Warehouse: Production design

Apply Data Warehouse through security, performance, operations.

LESSON 10 / 22 MIN

Security with ETL

Learning objectives

- Explain security in the context of Data Warehousing & BigQuery.
- Apply ETL to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

security, ETL, databases

Retrieval prompt

In Data Warehousing & BigQuery, which evidence best supports a security result produced with ETL?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 11 / 26 MIN

Performance with BigQuery

Learning objectives

- Explain performance in the context of Data Warehousing & BigQuery.
- Apply BigQuery to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

performance, BigQuery, databases

Retrieval prompt

In Data Warehousing & BigQuery, which evidence best supports a performance result produced with BigQuery?

Evidence to keep

☐ Original input ☐ Observed result ☐ Boundary or failure result ☐ Limitation and next action

LESSON 12 / 30 MIN

Operations with Snowflake

Learning objectives

- Explain operations in the context of Data Warehousing & BigQuery.
- Apply Snowflake to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

operations, Snowflake, databases

Retrieval prompt

In Data Warehousing & BigQuery, which evidence best supports a operations result produced with Snowflake?

Evidence to keep

[] Original input [] Observed result [] Boundary or failure result [] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable Data Warehousing & BigQuery project using BigQuery, Snowflake, dbt.

DELIVERABLE	A working Data Warehousing & BigQuery artefact plus an evidence-based self-review.
TOOLS	A suitable BigQuery environment, A plain-text decision log, Test data or realistic sample material

Production plan

- 1 Define the intended user, outcome and constraints.
- 2 Create the smallest complete result using BigQuery.
- 3 Test one normal case, one boundary case and one failure response.
- 4 Revise the work from the evidence and preserve before-and-after results.
- 5 Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER

State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS

List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE

Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE

What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION

What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION

Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

PostgreSQL Tutorial

PostgreSQL Global Development Group - reviewed 2026-08-21

<https://www.postgresql.org/docs/current/tutorial.html>

SQL Language

PostgreSQL Global Development Group - reviewed 2026-08-21

<https://www.postgresql.org/docs/current/sql.html>

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the BigQuery scope.